

Code-Layout, Readability and Reusability

Date	7 July 2026
Team ID	SWTID-2026-8977
Project Name	EduGenie Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Python PEP-8 indentation followed consistently.
2	Proper File Structure	Files and folders are logically organized	Yes	Project separated into frontend, backend, templates, static files, and API modules.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Variables such as <code>user_query</code> , <code>quiz_data</code> , <code>summary_text</code> , and <code>learning_path</code> improve readability.
4	Function / Method Names	Functions are descriptively named	Yes	Functions like <code>generate_quiz()</code> , <code>summarize_text()</code> , <code>answer_question()</code> , and <code>generate_learning_path()</code> clearly describe their purpose.
5	Code Comments	Inline and block comments explain logic	Yes	Important functions and API calls include comments for better understanding.
6	Modular Design	Code is split into reusable functions/modules	Yes	Separate modules are used for Gemini API integration,

				backend services, and frontend components.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Reusable helper functions reduce code duplication.
8	Error Handling	Exceptions and errors are handled gracefully	Yes	FastAPI exception handling and input validation ensure reliable execution.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Gemini API Integration Module	Python, Google Gemini API	Question Answering, Concept Explanation, Summarization	High
2	Quiz Generator	Python	Quiz Generation Feature	High
3	Learning Path Generator	Python	Personalized Learning Recommendation	High
4	FastAPI Backend Services	FastAPI	Handles all frontend API requests	High
5	HTML/CSS User Interface Components	HTML, CSS, JavaScript	Home Page, Results Page, Quiz Interface	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized project structure with separate frontend and backend modules.
Readability	5	Clear naming conventions, proper formatting, and easy-to-understand code.
Reusability	5	Modular functions and reusable API components minimize duplication.
Documentation / Comments	4	Sufficient comments provided for important modules and functions.
Overall Score	4.8/5	The project follows good coding standards with clean, modular, maintainable, and reusable code.